

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$x(t)$ } Periodic with period T and $y(t)$ } fundamental frequency $\omega_0 = 2\pi/T$	a_k b_k
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} x(t) = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a_{-k}^*
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t-\tau)d\tau$	$T a_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$j k \omega_0 a_k = j k \frac{2\pi}{T} a_k$
Integration		$\int_{-\infty}^t x(\tau) d\tau$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0} a_k - \left(\frac{1}{jk(2\pi/T)} a_k \right) a_k \right)$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$a_k = a_{-k}^*$ $\Re\{a_k\} = \Re\{a_{-k}\}$ $\Im\{a_k\} = -\Im\{a_{-k}\}$ $ a_k = a_{-k} $ $\angle a_k = -\angle a_{-k}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$x_e(t) = \mathcal{E}\{x(t)\}$ [x(t) real] $x_o(t) = \mathcal{O}\{x(t)\}$ [x(t) real]	$\Re\{a_k\}$ $j\Im\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

表 3.4 基本信号的卷积表

连续时间卷积积分			离散时间卷积和		
$x(t)$	$h(t)$	$x(t) * h(t)$	$x[n]$	$h[n]$	$x[n] * h[n]$
$x(t)$	$\delta(t)$	$x(t)$	$x[n]$	$\delta[n]$	$x[n]$
$x(t)$	$u(t)$	$\int_{-\infty}^t x(\tau) d\tau$	$x[n]$	$u[n]$	$\sum_{k=-\infty}^n x[k]$
$x(t)$	$\delta^*(t)$	$x^*(t)$	$x[n]$	$\Delta\delta[n]$	$x[n] - x[n-1]$
$x(t)$	$u(t)$	$tu(t)$	$x[n]$	$u[n]$	$(n+1)u[n]$
$e^{-at}u(t)$	$u(t)$	$\frac{1-e^{-at}}{a}u(t)$	$a^n u[n]$	$u[n]$	$\frac{1-a^{n+1}}{1-a}u[n]$
$\sin(\omega t)u(t)$	$u(t)$	$\frac{1-\cos(\omega t)}{\omega}u(t)$	$\sin(\omega n)u[n]$	$u[n]$	$u[n]$
$\cos(\omega t)u(t)$	$u(t)$	$\frac{\sin(\omega t)}{\omega}u(t)$	$\cos(\omega n)u[n]$	$u[n]$	$u[n]$
$e^{-at}u(t)$	$e^{-bt}u(t)$	$te^{-at}u(t)$	$a^n u[n]$	$a^n u[n]$	$(n+1)a^n u[n]$
$e^{-at}u(t)$	$e^{-bt}u(t)$	$\frac{e^{-at}-e^{-bt}}{b-a}u(t)$	$a^n u[n]$	$b^n u[n]$	$\frac{b^{n+1}-a^{n+1}}{b-a}u[n]$

TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(\alpha t)$	$\frac{1}{ \alpha } X\left(\frac{j\omega}{\alpha}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$X(j\omega) = X^*(-j\omega)$ $\Re\{X(j\omega)\} = \Re\{X(-j\omega)\}$ $\Im\{X(j\omega)\} = -\Im\{X(-j\omega)\}$ $ X(j\omega) = X(-j\omega) $ $\angle X(j\omega) = -\angle X(-j\omega)$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}\{x(t)\}$ [x(t) real] $x_o(t) = \mathcal{O}\{x(t)\}$ [x(t) real]	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals	$\int_{-\infty}^{+\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) ^2 d\omega$	

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$x[n]$ } Periodic with period N and $y[n]$ } fundamental frequency $\omega_0 = 2\pi/N$	a_k } Periodic with b_k } period N
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n} x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a_{-k}^*
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m} a_k$ (viewed as periodic with period mN)
Periodic Convolution	$\sum_{r=-\infty}^{+\infty} x[r]y[n-r]$	$N a_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)}) a_k$
Running Sum	$\sum_{k=-\infty}^n x[k]$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{1 - e^{-jk(2\pi/N)}} \right) a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$a_k = a_{-k}^*$ $\Re\{a_k\} = \Re\{a_{-k}\}$ $\Im\{a_k\} = -\Im\{a_{-k}\}$ $ a_k = a_{-k} $ $\angle a_k = -\angle a_{-k}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}\{x[n]\}$ [x[n] real] $x_o[n] = \mathcal{O}\{x[n]\}$ [x[n] real]	$\Re\{a_k\}$ $j\Im\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{N} \sum_{n=-\infty}^{+\infty} |x[n]|^2 = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

TABLE 1.1 Comparison of the signals $e^{j\omega_0 t}$ and $e^{j\omega_0 n}$.

$e^{j\omega_0 t}$	$e^{j\omega_0 n}$
Distinct signals for distinct values of ω_0	Identical signals for values of ω_0 separated by multiples of 2π
Periodic for any choice of ω_0	Periodic only if $\omega_0 = 2\pi m/N$ for some integers $N > 0$ and m .
Fundamental frequency ω_0	Fundamental frequency* ω_0/m
Fundamental period $\omega_0 = 0$: undefined $\omega_0 \neq 0$: $\frac{2\pi}{\omega_0}$	Fundamental period* $\omega_0 = 0$: undefined $\omega_0 \neq 0$: $m \left(\frac{2\pi}{\omega_0} \right)$

*Assumes that m and N do not have any factors in common.

TABLE 4.2 BASIC FOURIER TRANSFORM PAIRS

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_k = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_k = a_{-k} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_k = -a_{-k} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave $x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t+T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \delta(t - nT) \quad \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at}u(t), \Re\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at}u(t), \Re\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{e^{at}-1}{(a-j\omega)} e^{-at}u(t), \Re\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

